VISHWAS · SIH26079 · TEAM HELLFIRE LABS

Arnav

Forecast-failure case studies

Final prep — script, judge Q&A, and every number traced

BCA | Slide 2 first half, then Slide 4 second half | about 70 seconds on Slide 2, then about 40 more on Slide 4

YOUR JOB

You set up the whole argument: what VISHWAS is, and why the forecast underneath it cannot be trusted at the lead times that matter. You are also the person who answers 'give me a real example' — that is your research, and nobody else should take it. You now come back on after Paras to close Slide 4 with viability and the four risks.

1 RUNNING ORDER

Mahika leads. Your rows are highlighted.

#	SPEAKER	WHAT HAPPENS	TIME
1	Mahika	Slide 1 — opens. Team, problem statement, the hook.	~40s
2	**Arnav**	**Slide 2 — what VISHWAS is, and why the forecast underneath decays.**	**~70s**
3	Devansh	Slide 2 — the measurement, and the result.	~60s
4	Ankit	Slide 3 — opens it: what an ensemble is, and why we score against rain gauges.	~35s
5	Devansh	Slide 3 — the pipeline, the stack, features, label, model, the map.	~75s
6	Mahika	Bridge into slide 4.	~10s
7	Paras	Slide 4 — feasibility, the numbers, the three defects.	~65s
8	**Arnav**	**Slide 4 — viability, the four risks, the precision choice.**	**~40s**
9	Mahika	Slide 5 — the three interviews.	~45s
10	Dhruv	Slide 5 — who it serves, MoES's numbers, the benefits. Then slide 6 — sources.	~75s
11	Devansh	Slide 6 — our own measurement.	~20s
12	Mahika	Closes. The tagline, then into questions.	~20s

2 THE STORY — EVERYONE MUST BE ABLE TO SAY THIS

If a judge stops you in the corridor. Thirty seconds. Learn the shape, not the words.

Every weather forecast India issues arrives with no honesty label.

Forecasters judge confidence today by ensemble spread — run the model 11 times from slightly different starting conditions, and if the runs disagree, distrust the forecast. We measured that method against India's own rain gauges, across all four seasons and three years.

Compared crudely it looks strong: ensemble spread appears to explain 43.9% of the variance in forecast error. Compared fairly — within the same rainfall conditions — it explains under 1%.

So we built the replacement. On that same fair comparison it scores 0.565 and VISHWAS scores 0.728, on a test year the model had never seen.

Most teams pitch a hypothesis. We pitch a measurement.

3 WHAT CHANGED IN YOUR SCRIPT

Your wording is your own. Only these points moved.

- DECK CHANGED — your risk card was rewritten and your old line now contradicts it. The card used to say one test year is thin and that extending to five years needs no new code. That work is done: retrained on 2015 to 2017, rescored on 104 forecast dates across two monsoons, margin +0.149, whole interval above zero, ahead in every rainfall bin. Your script now says that. Quote the MARGIN and nothing else — never an AUC number beside it.
- LANGUAGE REBALANCED to roughly 70% English, 30% Hindi. Not one fact changed and the length is almost the same, nine words longer. The substance now lands in English and the Hindi carries the joins and the emphasis, because the panel is MoES and NCMRWF scientists and a technical argument delivered mostly in Hindi is the one that gets misheard. Read it aloud twice — it is your script, in different proportions.
- SPLIT — you now speak twice. After Paras finishes the numbers on Slide 4, you come back on for the bottom of that slide: viability, the four risk cards, and the precision design choice. About 40 seconds of new material — learn it as its own block, not as an extension of Slide 2.
- Two trims on Slide 2 to make room. Your opening asked the same question twice; there is one version now. And “every weather forecast in India arrives with no honesty label” is gone, because Mahika says that line about forty seconds before you — your “VISHWAS wohi honesty label hai” now picks hers up instead of repeating it, which is stronger.
- The risk questions are yours now — one test year, land-only, why GEFS and not NEPS, could this run at NCMRWF tomorrow. Anything with a number behind it still goes to Paras or Devansh.
- Your description of ensemble spread said the forecaster manually produces the forecast 11 times. It is the centre's computer that runs the model 11 times from slightly different starting conditions; what is manual is the forecaster's judgement of the result. This was the most serious error in any of the six scripts — and Ankit says it correctly ninety seconds after you, so the two of you were contradicting each other.
- “Forecast burst” corrected to “forecast bust.” It is the term in the problem statement title.
- 0.565 and 0.728 removed. They are Devansh's reveal on the very next card, and hearing them from you first flattens his moment. In their place you now have your own number — 23% at Day 1 falling to 0.2% at Day 10 — which nobody in the whole presentation was saying.
- “reliable ya unreliable” corrected to the three categories on Slide 2: reliable, uncertain, high-risk. Day 1 to Day 10 added.
- Your greeting is removed — Mahika greets for the team.
- The blue tick stays. It works.

4 YOUR SCRIPT

Grey boxes are stage directions — do not say them. Lines tagged TRIM are safe to cut if you are running long.

SLIDE 2 — FIRST HALF

▸ *Mahika hands to you: "Arnav will explain what VISHWAS is, and why today's forecasting system needs a reliability layer."*

Thank you, Mahika.

VISHWAS ka naam sunkar pehla sawaal yahi aata hai — is this another forecast model?

The answer is no.

▸ *Mahika has already said "every forecast arrives with no honesty label." Do not say it again — pick it up. "Wohi" is doing that work.*

VISHWAS wohi honesty label hai. A reliability layer — a blue tick on the forecast, jo batata hai us par kitna bharosa kiya ja sakta hai.

It scores every rainfall forecast by location and by lead time, Day 1 to Day 10, as reliable, uncertain, or high-risk.

Ye forecast ko badalta nahi hai. It only tells you when the forecast is likely to be wrong.

▸ *PAUSE*

Toh sawaal ye hai — why is this needed?

Today a duty forecaster judges confidence by looking at ensemble spread. The forecast centre runs the model 11 times, har baar thodi alag starting conditions ke saath. Agar woh 11 runs agree karte hain, the forecast is trusted. If they disagree, it is not.

Lekin ye judgement poori tarah manual hai — on every forecast cycle, under time pressure.

And the forecast underneath gets weaker with every passing day. For heavy rain — 64.5 millimetres or more in a day — GEFS detects only 23% of events at Day 1. Aur Day 10 tak, sirf 0.2%.

▸ *BIG PAUSE — look at the judges. This is your number and it is the reason the project exists.*

Yehi woh window hai jisme district authorities apni planning karte hain.

VISHWAS automates that judgement — aur usse ek evidence base deta hai.

Ab Devansh aapko batayenge ki humne us evidence ko kaise measure kiya.

▸ *Stop there. That last line is Devansh's cue — do not add anything after it.*

SLIDE 4 — SECOND HALF · NEW, THIS WAS PARAS'S

▸ *You come back on after Paras. He hands to you: "Ab Arnav aapko viability aur risks batayenge." Step straight in — no thank-you, the room has heard three already. You are speaking to the bottom of the slide: the viability card, then the four risk cards.*

Viability first. The features come from ensemble fields that every operational centre already produces. It runs on CPU, GPU ki zaroorat nahi — ye ek layer hai, not a new forecasting system.

Ab risks, jo hum khud bata rahe hain. One test year was thin, isliye humne dobara test kiya — 2015 to 2017 par train, aur 104 forecast dates par rescore, do monsoon ke across. Margin +0.149, whole interval above zero, every rainfall bin mein aage. One year was already enough.

▸ *This is the card the judges are reading. Do not say the AUC — quote the MARGIN, +0.149, and nothing else. If pressed on the numbers behind it, hand to Paras.*

Domain land tak limited hai, because that is what the IMD gauge grid covers. Aur NEPS ka public archive nahi hai, isliye humne GEFS use kiya — access milte hi we move to NEPS.

Aur ek design choice. In Smart Automation a false alarm costs more than a miss, isliye humne precision ki taraf tune kiya hai — not maximum detection.

▸ *That last line is a theme answer, not a metrics answer. If a judge asks for the operating point, the alert budget, or any number behind it, hand it straight to Paras.*

Ab Mahika aapko batayengi ki humne field mein kya suna.

▸ *Hand back to Mahika — she takes Slide 5. Stop there.*

5 EVERY NUMBER YOU SAY, AND WHERE IT COMES FROM

If a judge challenges a figure, give the source — not a defence.

CLAIM	SOURCE / WHAT IT REALLY MEANS
Day 1 to Day 10	The problem statement asks for region-wise confidence across exactly that range.
23% at Day 1, 0.2% at Day 10	Probability of detection for rain ≥ 64.5 mm. This is the UNDERLYING FORECAST, not our model — keep those separate and say so if pressed.
'flags busts, does not adjust the forecast'	A hard project rule. Detection, not correction.
1,252 of 5,041 (24.8%)	The IMD gauge domain. Say the fraction, not just "land only" — volunteering the number is what makes it a stated limit rather than a caught one.
CPU, no GPU	50.3 GB and 1.3 million gridpoint-days on a laptop. If pushed for the figures, they are Devansh's.
precision over detection	A design choice tied to the theme, NOT a metric. Any question about the operating point or the alert budget goes to Paras.

6 QUESTIONS THE JUDGES WILL ASK YOU

Yours to answer. Read them out loud once — an answer you have said aloud comes back under pressure; one you have only read does not.

Q1 Why GEFS and not NCMRWF's own NEPS?

Because NEPS reforecast archives are not public. GEFS version 12 is the only medium-range ensemble with a public twenty-year reforecast, and you need that history to learn what forecast error looks like.

Nothing in the method is specific to GEFS. It reads ensemble mean and spread — every centre produces both. The day we have NEPS access, we move.

⚠ Do not claim we have tested it on NEPS. We have not.

Q2 One test year is thin.

Agreed, and we say so on the slide rather than waiting to be asked. The interval resamples 52 dates inside a single year, so it is blind to year-to-year variation — treat it as a floor, not a ceiling.

Extending to five years needs no new code and no re-downloading.

⚠ If they push on what the interval actually is, that is Paras.

Q3 Why land only?

Because the truth is IMD's gauge-gridded rainfall, and gauges are on land. That is 1,252 of 5,041 gridpoints, and we state the fraction ourselves.

We could have scored against a reanalysis and covered the ocean, but a reanalysis shares model physics with the forecast being judged. We chose the harder, honest target.

⚠ The reanalysis point is Ankit's ground — if it goes deeper, hand it to him.

Q4 Could this actually run at NCMRWF tomorrow?

The inputs are ensemble fields they already produce every cycle. It is CPU only, no GPU. It is a layer over an existing run, not a new run.

What it needs from them is access to their own archive. Nothing else.

⚠ Any question about compute cost or data volume goes to Devansh.

Q5 Give me a real example of a forecast bust.

Mumbai, 26 July 2005. Santacruz recorded 944 millimetres in 24 hours. Colaba, 27 kilometres away, recorded 73. Same city, same day, an order of magnitude apart.

That is from Kumar and colleagues, 2008, in the Quarterly Journal of the Royal Meteorological Society.

It is the cleanest justification for why confidence has to be per gridpoint rather than per region. A single confidence number for ‘Mumbai’ that day would have been wrong for one of those two stations no matter what it said.

⚠ Our own earlier chart said Colaba 70 mm. The peer-reviewed figure is 73. Say 73.

Q6 What about Kerala, August 2018?

We looked at it and deliberately left it off the deck. Ashrit and colleagues, 2020, from NCMRWF, studies that event directly and is the right source.

What we could not do was source a specific claim about what IMD forecast in advance. Our internal draft had a figure for it that we could not trace to any published table — and we are presenting to that ministry — so we took it off. IMD's own report puts observed August 2018 Kerala rainfall at about 94% above normal.

⚠ Do not quote '+13% forecast vs +96% observed'. The 96% is wrong (it is ~94%) and the 13% has no citation.

Q7 How bad is the underlying forecast, really?

At the heavy-rain threshold — 64.5 millimetres a day — GEFS probability of detection is 0.060 overall, and detection falls from 23% at Day 1 to 0.2% at Day 10.

I would keep that clearly separate from our own scores: that is the forecast, not our model. Our job is not to fix that number. It is to tell the forecaster which of those forecasts to distrust.

Q8 **Why does ensemble spread fail?**

Because it mostly measures geography. Wet regions have both large spread and large errors, so pooled across India spread looks like it explains 43.9% of the variance in forecast error.

Compare forecasts in the same rainfall conditions and it drops under 1%. It is tracking how hard it rains, not how uncertain the forecast is.

Q9 **What is the human cost of a bust?**

We asked people directly. A farmer in Assam shifted a sowing date for rain that never arrived and lost seed, labour and time. An NDRF constable in Guwahati has mobilised for forecast heavy rain that did not come.

That is Mahika's field research and she can take you through it properly — but the pattern is that a bust in either direction costs somebody something.

Q10 **Isn't an extreme rainfall event the same as a forecast bust?**

No, and we were careful about the difference. There is an extreme-rainfall event we studied and deliberately left off the deck, because it documents extreme weather rather than a documented forecast failure.

A bust is the forecast being wrong. That is a different thing from the weather being extreme, and conflating them would have weakened the argument.

⚠ Do NOT quote rainfall figures for that case. It is Uttarakhand, June 2013, and our own verification record marks it 'not used and not verified'. Quoting unverified numbers in order to demonstrate sourcing discipline is the worst possible place to be caught. Name the principle, not the case.

Q11 **Why 64.5 millimetres?**

It is IMD's own heavy-rainfall threshold, so it is the operationally meaningful cut rather than one we picked.

Q12 **Does your system apply to cyclones and heat waves, or only rainfall?**

The problem statement names monsoon depressions, heavy rainfall, western disturbances, cyclones, heat waves and break-active monsoon phases. We built and measured on rainfall, because IMD's gauge-gridded rainfall is the truth source we can verify against.

The method is not rainfall-specific — it learns which ensemble disagreements matter for a given error definition — but I would not claim results for a variable we have not measured.

7 IF IT IS NOT YOURS, HAND IT OVER

Naming the right teammate reads as a team that knows its own work. Guessing reads as one that does not.

IF THEY ASK ABOUT…	SAY THE NAME
Any question about the model's score or how it was measured	Devansh
Metrics, statistics, leakage, the three defects, the alert budget	Paras
Where the data came from, the pipeline, the fetch	Devansh
What an ensemble is, why gauges rather than reanalysis	Ankit

IF THEY ASK ABOUT…	SAY THE NAME
The interviews in detail	Mahika
Sources and impact	Dhruv

SITUATION	WHAT TO SAY
You don't know the answer	“I don't have that number to hand and I'd rather not guess — can I come back to you on it?” That is a strength for this team, not a weakness: when we could not trace one of our own figures, we removed it rather than present it.
It's a teammate's area	“That's <name>'s part — <name>, do you want to take that?” Say the name. Don't answer for them.
A judge challenges a number	Give the source, not a defence. Every figure on the deck traces to a government source, a peer-reviewed paper, or our own logs.
A judge says 'you're just predicting heavy rain'	Anyone can answer this one: we removed the rainfall forecast from the model entirely and the score went UP, 0.728 to 0.731.

8 NEVER SAY THIS

Every one of these was a real error caught during the build, or a rule set deliberately.

- × Never say '43.9% of forecast error'. It is 43.9% of the VARIANCE in forecast error. This is the single easiest thing to get wrong under pressure.
- × Never quote the margin as '+0.163'. Say '+0.13 to +0.16'.
- × Never say 'deciles' or 'the wettest tenth' about the model result. It is EIGHT bins. (Deciles are correct only for the separate Phase 1 correlation analysis.)
- × Never quote any accuracy figure. Base rate is about 6%, so accuracy is meaningless — and quoting it signals a team that does not understand imbalance.
- × Never say 'AI-adjusted forecast' or imply we correct the forecast. We DETECT busts. We do not correct anything.
- × Never volunteer the pooled ROC-AUC pair (0.909 vs 0.885). It is a Q&A-only answer if a judge asks directly what the crude comparison gives. Putting it forward would commit, in our own work, exactly the error our pitch accuses the incumbent of.
- × Never quote +0.663 as 'the baseline we beat', and never quote a margin range containing +0.035 (that one came from a filter conditioning on the outcome).
- × Never call the regions 'IMD official regions'. They are approximate lat/lon boxes.
- × Never present the confidence map's date as evidence of general performance. It is ONE date, illustrating the form of the product.
- × Never convert the three field interviews into percentages or 'N% of users said'. They are three exploratory conversations.
- × Never quote 98%→46%, ₹13,331 cr, '50% faster', '~30% improvement', or a 0.34–0.37 false-alarm ratio. All were in earlier drafts; none could be traced; all were removed. Do not say any of these numbers out loud even while explaining that you removed them.
- × Never say SHAP and permutation importance 'agree on the drivers'. They pick the same top three in a different order.

9 NUMBER CHEAT SHEET

Same in all six packs. If two of you quote a figure differently in the same room, that is the damage.

WHAT	NUMBER	WATCH OUT
Spread explains, pooled	43.9% of the VARIANCE in forecast error	not '43.9% of forecast error'
Spread explains, fair comparison	0.17%–0.90% → say 'under 1%'	never quote +0.041 alone
VISHWAS score (stratified ROC-AUC)	0.728	—
Ensemble spread score	0.565	—
Margin	+0.13 to +0.16 · 95% CI +0.108 to +0.193	never '+0.163' alone
Matched 10% alert budget	VISHWAS catches 64% of busts vs 55%	—
POD / FAR / CSI / ETS at 10%	0.641 / 0.586 / 0.336 / 0.300 vs 0.546 / 0.647 / 0.273 / 0.235	—
Features	23, all from GEFS, none from the truth	not 24
Ensemble members	11 (control + 10 perturbed)	—
Domain	1,252 of 5,041 gridpoints (24.8%), land only	—
Data fetched	50.3 GB · 104 forecast dates	—
Labelled rows	1,302,080 total · 651,040 in the test set	don't conflate the two
Train / test	trained on 2018 · tested on all 52 Wednesdays of 2019	—
Base rate	about 6%	so never quote accuracy
Bust definition	|forecast − observed| above the 95th percentile, per region and lead time, threshold set on training data only	—
Rainfall bins	EIGHT bins · 40.92% bust rate in the wettest, 0.15% in the driest	never 'deciles' or 'wettest tenth'
Ablations	full 0.728 · no month 0.722 · no rainfall forecast 0.731 · no 850 shear 0.726 · spread-only 0.562	—
Underlying GEFS, heavy rain (≥64.5 mm)	detects 23% of events at Day 1, 0.2% at Day 10	this is the forecast, NOT our model
IMD warning accuracy (MoES)	85, 73, 67, 63, 58% for Day 1–5	Lok Sabha Q.2953, 17 Dec 2025
The tie	ensemble spread is exactly 0.00 mm on 189,758 test rows — a third of the set	—
Calibration	Brier 0.04217 vs 0.04749 · resolution 1.4× the incumbent's	—

ONE LAST THING

The strongest thing about this project is not the score. It is that we measured the method used today before proposing a replacement, and that we removed our own figures when we could not source them. If you land one impression, land that one.